

## CS 673 : Advanced Computer Vision

Course Code : CS 673

Course Name : Advanced Computer Vision

L-T-P-C : 3-0-2-4

Intended for : B.tech (Yd and 4111 Year), CSE, DSE and EE Students

Prerequisite : EE511 Computer Vision or CS671 Deep Learning or EE608 Digital Image Processing or CS669 Pattern Recognition, or similar

Mutual Exclusion:

Approval: 54th BoA

### Course Contents

- **Introduction to Computer Vision:** Get a conceptual overview of image classification, object localization, object detection, and image segmentation. Also be able to describe multi-label classification and distinguish between semantic segmentation and instance segmentation. In the rest of this course, you will apply PyTorch to build object detection and image segmentation models. (6 Hours)
- **Vision Models Classification:** Convolutional Neural Networks: Architectures, Convolution I Pooling Layers, spatial arrangement, layer patterns, layer sizing patterns, AlexNet/ZFNet/VGGNet/EfficientNet case studies, computational considerations, Transfer learning for Representation, domain adaptation, domain Generalization, open set, open world, curriculum learning (6 Hours)
- **Vision Models Object Detection and Tracking:** Overview of some popular object detection models, such as regional-CNN and ResNet-50. Use of object detection models in PyTorch, Transfer Learning for object detection: download your own models and configure them for training and build your own models for object detection. Using transfer learning, you will train a model to detect and localize. Model compression and real-time inference on Edge-devices. (9 Hours)
- **Vision Models Segmentation:** Types of semantic segmentation, instance-based models, Disadvantages of search window methods, RCNN-family models, Unet, DeepLab, Transformer based segmentation. (6 Hours)
- **Graphs and XAI for Vision:** Graph Representation/formulation for visual tasks, Deep Learning Models for Graphs, GradCam, GradCam++, Graph Model Explanation, LIME, GraphLIME (6 Hours)
- **Vision & Language Models:** Visual to Text: Recent methods for Text embeddings, Scene/Video Captioning, OCR, VQA, Video Description using RNNs, LSTMs. Transformers. Text to Visual: Text to image/video generation using V AEs, GANs, Transformers. (9 Hours)

### Textbooks:

1. Richard Szeliski, **Computer Vision: Algorithms and Applications**.
2. David Forsyth and Jean Ponce, **Computer Vision: A Modern Approach**, 2nd Edition.

### References:

1. Ian Goodfellow, **Deep Learning**.



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2. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, **Elements of Statistical Learning**.
  3. Richard Hartley and Andrew Zisserman, **Multiple View Geometry in Computer Vision**, 2nd Edition.